

Two-population classification

Applications

- Pattern recognition / detection
 - Machine learning
 - Digital communication / detection of signals
 - Medicine / pathology
 - Feature extraction
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Training data

- already collected “ground truth” data with known/defined population membership

New data

- “real” new observations for which definitive establishment of population membership would be
 - too expensive
 - too slow
 - impossible
 - only possible by destructive test
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Summarizing two-population discriminant analysis procedures

Training data

- collect “ground truth” data
- define or estimate prior probabilities, $p_i = P(\pi_i)$, $i = 1, 2$
- define misclassification costs, $C(1|2)$, $C(2|1)$
- estimate parameters for $f(\mathbf{x}|\pi_i)$, i.e. $\hat{\mu}_i, \hat{\Sigma}_i$, $i = 1, 2$
- MVN model control
- test $H_0: \Sigma_1 = \Sigma_2 \ggg$ LDA case for accept / QDA case for reject
- test $H_0: \mu_1 = \mu_2 \ggg$ significant separation of populations for reject
- specify/calculate sample based discriminant functions for LDA or QDA
- calculate confusion matrix for training data using the discriminant function
- calculate estimated classification error rates, APER and/or AER_{hat}

New data

- observe new data $\mathbf{x}_0$
 - classify $\mathbf{x}_0$ as member of population 1 or 2 using the calculated discriminant functions for the training data
 - calculate posterior probabilities using
 - $P(\pi_1|\mathbf{x}_0) \propto p_1 f(\mathbf{x}_0|\pi_1)$ (posterior₁ $\propto$ prior₁·likelihood₁)
 - $P(\pi_2|\mathbf{x}_0) \propto p_2 f(\mathbf{x}_0|\pi_2)$ (posterior₂ $\propto$ prior₂·likelihood₂)
 - $P(\pi_1|\mathbf{x}_0) + P(\pi_2|\mathbf{x}_0) = 1$ (normalization to actual probabilities)
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